

Special topics in logic and security II *

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1. The code C consists of the words (x, y, z) of length 3 over the alphabet $\mathbb{F}_{11}$ that satisfy an equation:

$$ax + by + cz = 0,$$

where $a, b, c \in \mathbb{F}_{11} \setminus \{0\}$ are fixed parameters. Find the weight polynomial for the code C .

2. Let $C^\perp$ be the orthogonal complement of the code defined in the previous exercise. Apply MacWilliams' formula to compute the weight polynomial of $C^\perp$ and explain the result.
3. Consider the code:

$$G = \{(P(1), P(2), P(3), P(4), P(5), P(6)) \in \mathbb{F}_{11}^6 \mid P \in \mathbb{F}_{11}[X], \deg(P) \leq 2\}.$$

Show that $w = (6, 6, 1, 2, 9, 0) \in G$.

4. Which of the following matrices is unitary? Prove the answer!

$$R_1 = \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}, \quad R_2 = \frac{1}{2} \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}.$$

5. Prove or justify the following matrix identity:

$$(A \otimes B)(C \otimes D) = AC \otimes BD.$$

6. Write down the discrete Fourier transform $\widehat{f}(x)$ for a function $f : \mathbb{Z}_4 \rightarrow \mathbb{C}$ as an explicit sum of four terms.

*Master, 4-th semester. IF